

MATH 189 HW 2

Dani Nguyen

Winter 2025

Question 1

- a. Since IQ and GPA is fixed, we will denote those values as a constant C

$$\begin{aligned} Y &= C + 35X_3 - 10X_5 \\ &= C + \mathbb{I}\{\text{has college}\}(35 - 10X_5) \end{aligned}$$

When fixing IQ and GPA, the model predicts that college graduates will earn less if their GPA is higher than 3.5. Therefore, option iii. is correct.

- b. For a college graduate with IQ 110 and GPA 4.0,

$$\begin{aligned} Y &= 50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(4.0 \cdot 110) - 10(4.0 \cdot 1) \\ &= 137.1 \text{ thousand dollars} \end{aligned}$$

- c. False, because GPA and IQ are on average very large numbers, especially when multiplied, that even a low coefficient can lead to large effects in salary. This does not mean that there is little evidence of an interaction effect, one would need to look at the p -values of the interaction term to know whether there exists an interaction effect.

Question 2

$$\begin{aligned}\hat{y}_i &= x_i \hat{\beta} \\ &= x_i \frac{\sum_{i=1}^n x_i y_i}{\sum_{i'=1}^n x_{i'}^2} \\ &= \frac{\sum_{i=1}^n x_i^2 y_i}{\sum_{i'=1}^n x_{i'}^2} \\ &= \sum_{i=1}^n \frac{x_i^2}{\sum_{i'=1}^n x_{i'}^2} y_i\end{aligned}$$

Let $a_i = \frac{x_i^2}{\sum_{i'=1}^n x_{i'}^2}$

$$\hat{y}_i = \sum_{i=1}^n a_i y_i$$

Question 3

Recall that $\beta_0 = \bar{y} + \beta_1 \bar{x}$. This can be rearranged to $\bar{y} = \beta_0 - \beta_1 \bar{x}$. This means that the equation for simple linear regression will always pass through the point $(\bar{x}, \bar{y})$.

Question 4

$$\begin{aligned}R^2 &= \frac{TSS - RSS}{TSS} \\ &= \sum_{i=1}^n (y_i)^2 Cor(x, y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}\end{aligned}$$